Shayan Dodge

PhD Researcher | Scientific Machine Learning and AI for Physics

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Research Profile

PhD researcher in computational electromagnetics and scientific machine learning at the University of Pisa. My research develops physics-informed neural networks, variational methods, and neural operators for electromagnetic partial differential equations, with particular emphasis on systems that include nonlinear materials, material interfaces, and parametric simulation. My current research focuses on physically consistent operator learning and computationally efficient electromagnetic simulation.

Research Highlights

- Research experience in Scientific Machine Learning, Physics-Informed Neural Networks (PINNs), Neural Operators, GPU-Accelerated Computing, and Computational Electromagnetics.
- **21 peer-reviewed publications**: 18 journal articles and 3 conference papers.
- Developed **INI-VPINN** and **STAR-PINN**; contributed to physics-informed DeepONet and hybrid BEM–PINN methods for electromagnetic problems.
- Recipient of the **2026 IEEE Antennas and Propagation Society (AP-S) Doctoral Research Grant** (US$4,000; 10 recipients selected from 72 eligible applicants).
- Recipient of a **2024 University of Pisa research grant for machine learning in electromagnetics**; active collaborations with KU Leuven and Italian research institutions.
- **Student Editorial Assistant**, *IEEE Journal on Multiscale and Multiphysics Computational Techniques* (2025–Present).

Education

Ph.D. in Electrical Engineering, University of Pisa (DESTEC), Italy — Oct 2024–Present
(*Research Focus: AI in Computational Electromagnetics.*)

M.Sc. in Plasma Physics, Shahid Beheshti University, Iran — Sep 2017–Feb 2020
(WES-evaluated U.S. GPA: **3.87/4.00**)

B.Sc. in Physics, Kharazmi University, Iran — Oct 2013–Jun 2017
(WES-evaluated U.S. GPA: **3.98/4.00**)

Research Experience and Funded Projects

Research Fellow – University of Pisa (DESTEC), Italy — Jan 2024–Present
Conducting research in AI-driven computational electromagnetics and scientific machine learning through nationally and regionally funded projects in collaboration with academic and industrial partners in Italy.

AIGEO – AI for Electromagnetic Geophysical Inversion — Aug 2026–Present
Partners: University of Pisa (DESTEC); Geostudi Astier (geophysical engineering company)
Funding: European Regional Development Fund (ERDF/FESR), Regione Toscana.

- Developing data-driven and physics-informed neural networks for electromagnetic geophysical inverse problems, particularly Electrical Resistivity Tomography (ERT).

FELINES – AI for Lightning Detection and Power-System Protection — Jan 2024–Jan 2026
Partners: University of Pisa; University of Campania; University of Genoa
Funding: PRIN (Research Projects of National Interest), Italian Ministry of University and Research (MUR).

- Developed deep-learning models for lightning localization, peak-current estimation, and lightning-induced overvoltage prediction. [Website] [GitHub]

STEM-DEEP – Deep Learning for Personalized TMS — Jan 2024–Jan 2026
Partners: University of Pisa; University of Pavia; Politecnico di Milano; IRCCS Fondazione Mondino (neurological hospital)
Funding: PRIN (Research Projects of National Interest), Italian Ministry of University and Research (MUR).

- Developed generative deep-learning models combining variational autoencoders and CNN-based inverse modeling for patient-specific transcranial magnetic stimulation (TMS) and coil-position optimization. [Website] [GitHub]

Other Research Projects and Collaborations

Physics-Informed Neural Operators for Parametric Electromagnetic Modeling Apr 2025–Present
University of Pisa; KU Leuven, Belgium

- Contributed to physics-informed DeepONet models for parametric magnetostatic and magnetoquasistatic devices, including nonlinear materials and time-dependent problems. [Paper]
- Investigating physics-informed Fourier neural operators (PI-FNO) for nonlinear time-domain magnetoquasistatic simulations (manuscript under review).

Physics-Informed and Variational Neural Solvers for Electromagnetic PDEs Jul 2024–Present
University of Pisa; University of Campania

- Developed **INI-VPINN**, a variational formulation with implicit enforcement of Neumann and material-interface conditions in multi-material domains, including geometric singularities. [Paper] [GitHub]
- Developed **STAR-PINN**, using stacked adaptive residual learning for nonlinear time-dependent magnetic diffusion. [Paper] [GitHub]
- Contributed to hybrid BEM–PINN methods and mapped and non-conforming variational formulations for electromagnetic problems in complex geometries. [Paper] [GitHub]

Deep Learning for Real-Time SAR Assessment Sep 2024–Mar 2025
University of Pisa; University of Pavia

- Developed deep-learning surrogate models for rapid prediction of specific absorption rate (SAR) hot spots induced by wearable electromagnetic devices, supporting efficient exposure assessment. [Paper]

GPU-Accelerated FDTD for Plasma and Microwave Applications Sep 2018–Jan 2024
Shahid Beheshti University

- Developed GPU-accelerated FDTD simulations for electromagnetic wave propagation in plasma. [Preprint]

Technical Skills

Scientific ML: PINNs/VPINNs, DeepONet, Fourier neural operators, surrogate modeling, inverse problems; PyTorch, JAX, TensorFlow.
GPU and programming: CUDA, GPU-accelerated FDTD, parallel computing, Python, C++, MATLAB.
Numerical methods: FEM, FDTD, BEM, nonlinear electromagnetic PDEs; FEniCS, Gmsh, ANSYS Maxwell, CST Studio Suite, SimNIBS.
Software: Linux, Git, scientific software development.

Selected Publications

1. **Dodge, S.**, Barmada, S., and Formisano, A. (2026). INI-VPINN: A Variational Physics-Informed Neural Network with Implicit Neumann and Interface Handling for Multi-Material Domains with Geometric Singularities. *Journal of Computational Physics*. [Paper] [GitHub]
2. Toghranegar, S., **Dodge, S.**, Barmada, S., et al. (2026). Physics-Informed Deep Operator Networks for Parametric Modelling of Electromagnetic Devices: Magnetostatics to Magnetoquasistatics. *IEEE Transactions on Magnetics*. [Paper]
3. **Dodge, S.**, Barmada, S., and Formisano, A. (2025). A STacked Adaptive Residual PINN (STAR-PINN) Approach to 2D Time-Domain Magnetic Diffusion in Nonlinear Materials. *IEEE Access*. [Paper] [GitHub]
4. Barmada, S., **Dodge, S.**, Tucci, M., et al. (2024). A Novel Hybrid Boundary Element–Physics Informed Neural Network Method for Numerical Solutions in Electromagnetics. *IEEE Access*. [Paper] [GitHub]

Journal Publications

1. **Dodge, S.**, Toghranegar, S., Formisano, A., Sabariego, R. V., & Barmada, S. (2026). Physics-Informed Fourier Neural Operators (PI-FNO) for Time-Domain Magnetoquasistatic Analysis of Nonlinear Electromagnetic Devices. *IEEE Journal on Multiscale and Multiphysics Computational Techniques*. [Paper]
2. Barmada, S., **Dodge, S.**, & Formisano, A. (2026). Isoparametric Mapped Weak Form PINNs for Arbitrary Quadrilateral Elements vs. Unfitted Domain Approach. *IEEE Transactions on Magnetics*. [Paper]
3. **Dodge, S.**, Mestriner, D., Nicora, M., Barmada, S., Brignone, M., Formisano, A., & Procopio, R. (2026). Prediction of lightning flashover on overhead distribution lines from preliminary breakdown voltages based on deep learning. *Electric Power Systems Research*. [Paper]
4. Hajian, M., **Dodge, S.**, Fontana, N., & Barmada, S. (2026). Fast parametric prediction of coil coupling in wireless power transfer systems using physics-informed DeepONet. *IEEE Transactions on Magnetics*. [Paper]

5. **Dodge, S.**, Barmada, S., & Formisano, A. (2026). INI-VPINN: A Variational Physics-Informed Neural Network with Implicit Neumann and Interface Handling for Multi-Material Domains with Geometric Singularities. *Journal of Computational Physics*. [Paper]
6. Toghranegar, S., **Dodge, S.**, Barmada, et al. (2026). Physics-Informed Deep Operator Networks for Parametric Modelling of Electromagnetic Devices: Magnetostatics to Magnetoquasistatics. *IEEE Transactions on Magnetics*. [Paper]
7. Barmada, S., **Dodge, S.**, Formisano, A., Di Barba, P., & Mognaschi, M. E. (2026). PINN-Based Resolution of Inverse Non-linear Magnetostatic Problems. *COMPEL*. [Paper]
8. Codecasa, L., Zhu, X., Di Rienzo, L., Barmada, S., **Dodge, S.**, et al. (2026). Fast Model Order Reduction Based Approach for Transcranial Magnetic Stimulation with Varying Coil Positioning. *IEEE Access*. [Paper]
9. Formisano, A., **Dodge, S.**, & Barmada, S. (2025). A Comparison of Machine Learning and Classical Numerical Approaches for the Resolution of Electromagnetics Problems. *IET Science, Measurement & Technology*, 19(1), e70034. [Paper]
10. Barmada, S., **Dodge, S.**, & Formisano, A. (2025). Weak Formulation for Physics-Informed Neural Networks in the Resolution of Analysis Problems in Electromagnetics. *IEEE Transactions on Magnetics*. [Paper]
11. Cosentino, G., Zaffina, C., Zoccola, C., Fresia, M., Merli, S., Mauramati, S., Bertino, G., Todisco, M., **Dodge, S.**, et al. (2025). Unilateral EMG-Guided Botulinum Toxin for Retrograde Cricopharyngeus Dysfunction. *Toxins*, 17(9), 458. [Paper]
12. Barmada, S., **Dodge, S.**, Brignone, M., et al. (2025). Relating transmission line overvoltages and lightning location: a machine learning-based procedure. *COMPEL*. [Paper]
13. **Dodge, S.**, Barmada, S., & Formisano, A. (2025). A STacked Adaptive Residual PINN Approach to 2D Time-Domain Magnetic Diffusion in Nonlinear Materials. *IEEE Access*. [Paper]
14. **Dodge, S.**, Fontana, N., Mognaschi, M. E., et al. (2025). A Deep Learning Based Prediction of Specific Absorption Rate Hot-Spots Induced by Broadband Electromagnetic Devices. *IET Science, Measurement & Technology*, 19(1), e70009. [Paper]
15. **Dodge, S.**, Nicora, M., Barmada, S., et al. (2025). A deep learning based lightning location system. *Electric Power Systems Research*, 242, 111437. [Paper]
16. Sekehravani, E. A., **Dodge, S.**, Barmada, S., et al. (2025). Preliminary Breakdown Pulses: A review on available data and models. *Electric Power Systems Research*, 242, 111463. [Paper]
17. Barmada, S., **Dodge, S.**, Tucci, M., et al. (2024). A Novel Hybrid Boundary Element–Physics Informed Neural Network Method for Numerical Solutions in Electromagnetics. *IEEE Access*. [Paper]
18. Niknam, A. R., **Dodge, S.**, Hajian, M., et al. (2024). Characterization of microwave heating for hyperthermia cancer treatment. *Waves in Random and Complex Media*, 34(1), 211–225. [Paper]

Manuscripts Under Review

1. **Dodge, S.**, Fontana, N. Tucci, M, Barmada, S., et al. (2026). Real-Time Patient-Specific TMS Coil Placement Optimization Using a Generative Deep Learning Framework. *Scientific Reports*. Under review.

Conference Abstracts Under Review

- **Dodge, S.**, Formisano, A., & Barmada, S. (2027). *Explicit to Fully Implicit Boundary Treatment in Variational Physics-Informed Neural Networks: hp-VPINN, INI-VPINN, and UVPINN*. European Conference on Scientific Machine Learning (ECSML 2027), Naples, Italy, February 8–10, 2027. **Abstract submitted.** [Conference]

Conference Papers

1. **Dodge, S.**, Hajian, M., & Barmada, S. (2026). A non-conforming domain-decomposed variational physics-informed neural network for electromagnetic problems in complex geometries. *ACES*(pp. 1-2). IEEE. [Paper]
2. Barmada, S., Bonfiglio, A., Brignone, M., **Dodge, S.**, Formisano, A., Procopio, R., & Akbari Sekehavani, E. (2025). A Survey of Measurements and Analyses of Preliminary Breakdown Pulses in Lightning Flashes. *EEEIC/I&CPS Europe*. [Paper]
3. Nicora, M., Procopio, R., Brignone, M., **Dodge, S.**, et al. (2025). A Deep Learning Model for Lightning Location and Peak Current Estimation from Induced Overvoltages. *SIPDA*. [Paper]

Preprint

- **Dodge, S.**, Shafiee, M., and Shokri, B. (2022). Application of GPU-accelerated FDTD method to electromagnetic wave propagation in plasma using MATLAB Parallel Processing Toolbox. *arXiv preprint*. [Paper]

Honors and Awards

- **IEEE Antennas and Propagation Society Doctoral Research Grant**, 2026. Selected among 10 recipients from 72 eligible applicants; US$4,000 research support.
- **Application of Machine Learning to Electromagnetism Problems**, research grant, University of Pisa (DESTEC), 2024. Funded through FELINES and STEM-DEEP projects.
- Ranked first among M.Sc. Plasma Physics graduates, Shahid Beheshti University, 2020; ranked first among undergraduate Physics students, Kharazmi University, 2017.
- Top 1.5% in Iran's national B.Sc. entrance exam (251,000 participants), 2013.

Teaching and Supervision

- Teaching Assistant, Electrical Circuit Theory, University of Pisa (2024/25, 2025/26, 2026/27).
- Co-supervised two B.Sc. biomedical engineering theses on FEM-generated TMS datasets and deep-learning modeling (2024/25–2025/26).

Academic Service

- **Student Editorial Assistant**, *IEEE Journal on Multiscale and Multiphysics Computational Techniques* [Editorial Board] (Dec 2025–Present);
- **Organizing Committee Member**, 19th International Workshop on Optimization and Inverse Problems in Electromagnetism (OIPE 2027), Pisa, Italy, September 5–9, 2027.
- Reviewer for seven academic journals, including *IEEE Transactions on Magnetics* and *IEEE Journal on Multiscale and Multiphysics Computational Techniques*, as well as journals published by Springer Nature.

Pisa, Italy, November 7, 2026



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